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Education

H.B.Sc., Astronomy & Astrophysics, Statistics, Mathematics 2018-2022
University of Toronto

Technical Skills

Data: Python, NumPy, SciPy, Pandas, PyTorch, Scikit-Learn, Matplotlib, Seaborn, SQL, NoSQL, R, Excel
Web: HTML5, CSS3, JavaScript, React (Native), Redux, Node, Express.js, Flask
Other: shell scripting, vim, version control (git), cloud computing (AWS, Azure, Google Cloud Platform), Unix/Linux, SymPy, Astropy, Selenium, NLTK, $\text{\LaTeX}$, regular expressions

Research Experience

2020 Summer **Summer Undergraduate Research Program**
Dunlap Institute for Astronomy and Astrophysics, University of Toronto
Supervised by Dr. Allison Man

Independent Coursework

- 2019 **Data-Driven Astronomy**, The University of Sydney (Coursera)
Pulsars, black holes, exoplanets, stellar evolution, distances, redshift, galaxy classification. Image stacking, algorithm analysis, crossmatching, databases, regression, classification, machine learning. Implementation in Python with Astropy; SQL.
- 2019 **Intro to Deep Learning with PyTorch**, Udacity
Mathematics of neural networks. Data preparation, regularization techniques. Image classification with CNNs and OpenCV. Time series with RNNs. Implementation in Python with PyTorch.
- 2018 **Intro to Machine Learning**, Udacity
Classifiers (naïve Bayes, SVMs, ensemble methods), data processing, feature selection, regression, clustering, outlier detection, PCA, evaluation metrics. Implementation in Python with scikit-learn.

Projects

- 2019 **Image classification with CNNs**
Cleaned a dataset of several thousands of images of flowers. Optimized a deep learning model in Python and applied cutting-edge techniques to achieve 99% accuracy in the classification of species of flowers. Could potentially be adapted for classification of astronomical objects.
- 2018 **Analysis of global Internet usage statistics**
Spearheaded the programming aspect of a collaborative statistical analysis project. Proposed research questions and used R to develop data-backed answers. Gave a presentation of results.

Outreach

- 2019 - Present **Space Science Journalist (Current in Space segment)**, *The Star Spot* podcast
Write and record research-focused stories about the latest news in astronomy.

Volunteering

- 2015 - 2018 **Python Workshop Lead**, Sir Winston Churchill Secondary, Vancouver, B.C.
Coordinated and delivered workshops to prepare students for national programming contest.

Relevant Coursework

2020 Winter	Galaxies and Cosmology
2020 Winter	Practical Physics I
2020 Winter	Electricity and Magnetism
2020 Winter	Thermal Physics
2019 - 2020	Probability and Statistics I, II
2019 Fall	Stars and Planets
2019 Fall	Ordinary Differential Equations
2019 Summer	Multivariable Calculus
2019 Winter	Introduction to Computer Science
2018 - 2019	Calculus
2018 - 2019	Linear Algebra I, II
2018 Fall	An Introduction to Statistical Reasoning and Data Science

Interests

Languages: English (native), French, Chinese, Norwegian (basic)

Other: *nix customization, 3D modelling, urban beekeeping, baking, reading